Voice Diaries as Digital Biomarkers: A Scoping Review

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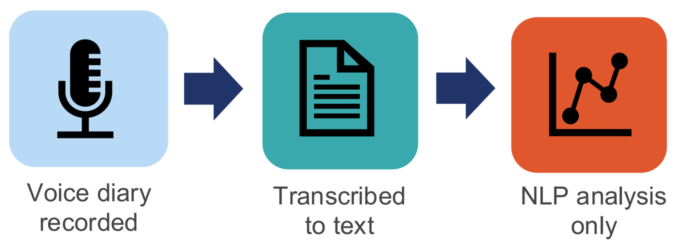
Voice Diaries in Health Research

- Voice diaries capture **spontaneous, ecologically valid speech** in daily life with low participant burden [1].
- Speech biomarkers show diagnostic and prognostic value, but rely on **constrained tasks** in controlled settings
- Modern platforms enable automated reminders, secure uploads, and real-time transcription.
- These advances position voice diaries as scalable tools for digital biomarker discovery, but computational potential of voice diaries **remains largely untapped.**

The Current State

Acoustic signals discarded. Pitch, voice quality, volume, all lost in transcription, carry clinically meaningful information [2].



Aim

To **map how voice diaries are used as digital biomarkers**, focusing on:

1. signal modalities and feature extraction approaches;
2. study designs, populations, and clinical targets; and
3. methodological gaps and translational challenges.

Methods

- **7 databases searched**: PubMed, Scopus, ACM Digital Library, IEEE Xplore, APA PsycInfo, Web of Science Core Collection, & ClinicalTrials.gov using the **Population-Concept-Context (PCC) framework**:
  (1) human participants providing intentional, longitudinal spoken entries;
  (2) computational extraction of acoustic, linguistic, paralinguistic features
  (3) focus on health or mental-health outcomes
- **Three reviewers screened studies** (κ = 0.82), and data were synthesized by application, context, and analytic approach.

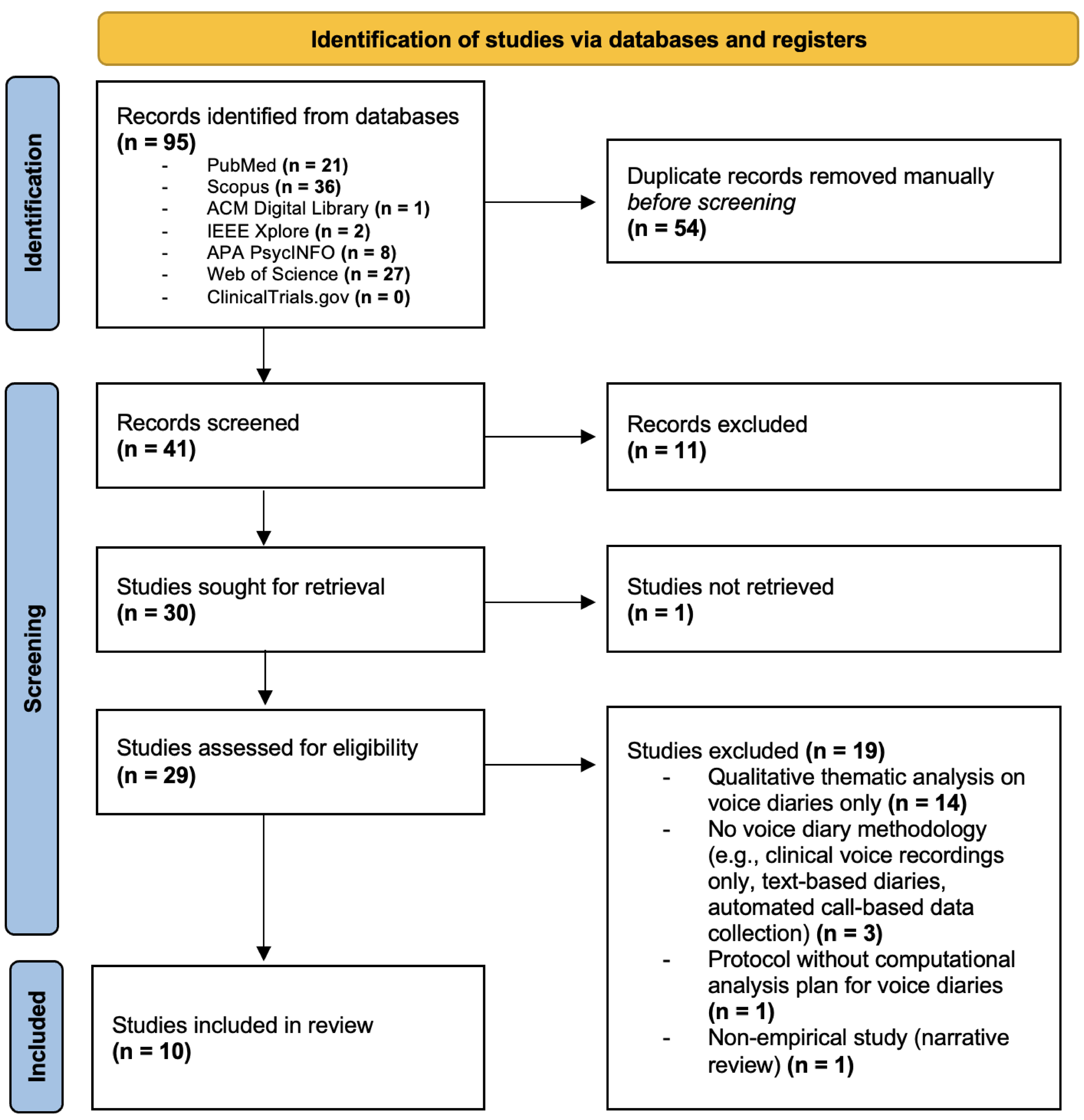
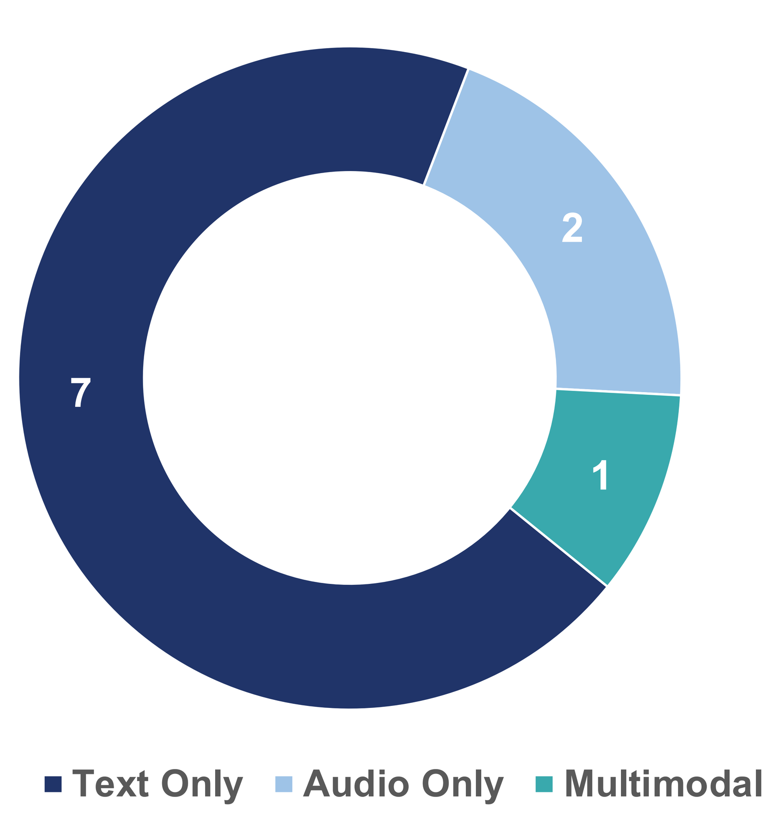
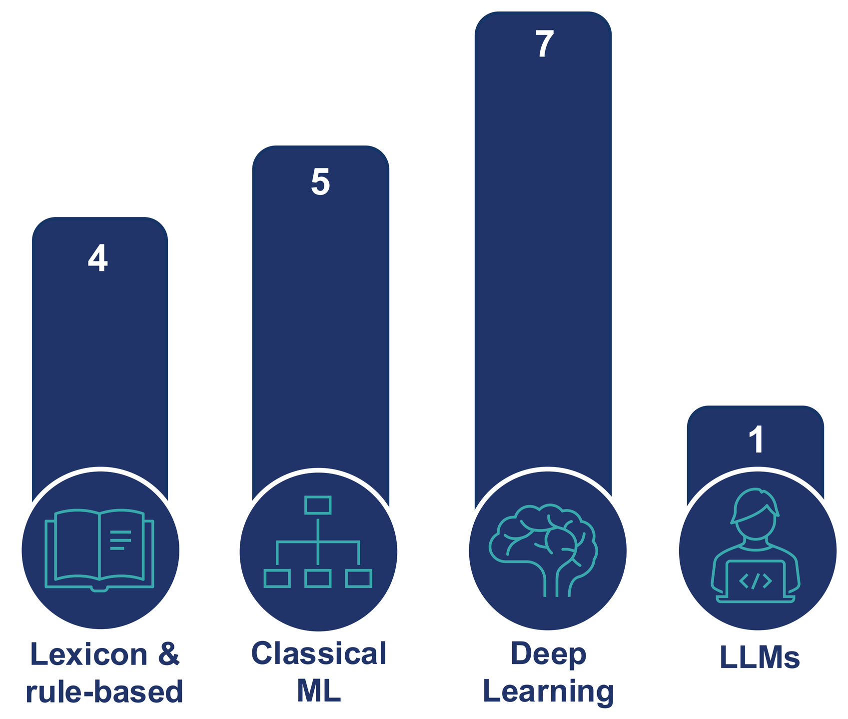


Figure 1. This review followed the PRISMA-ScR guideline The full protocol including search strategy and inclusion criteria is available on OSF [3].

Results

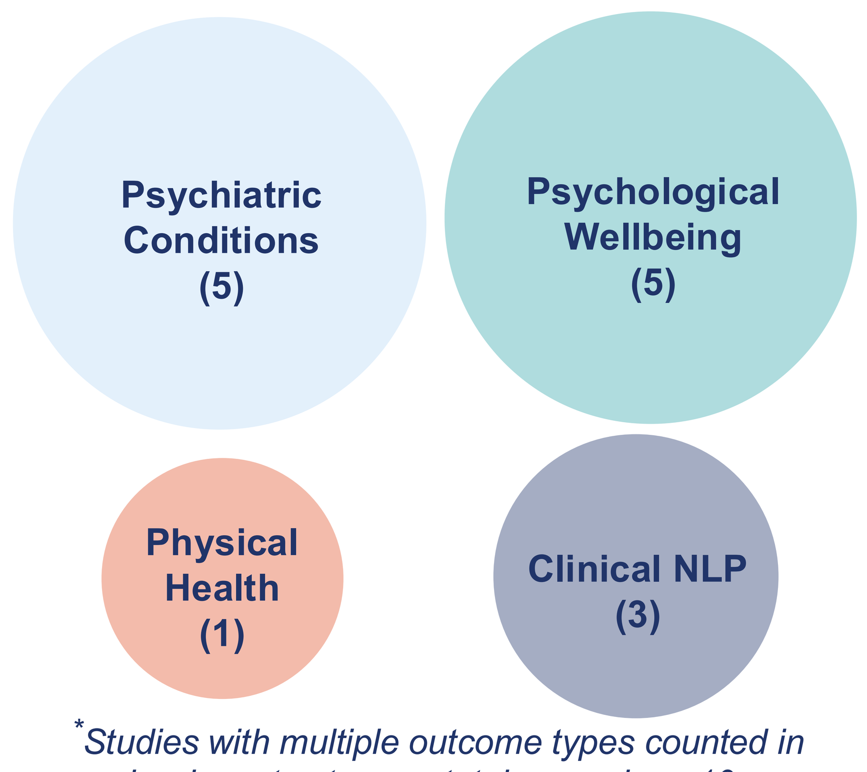


Signal Modality



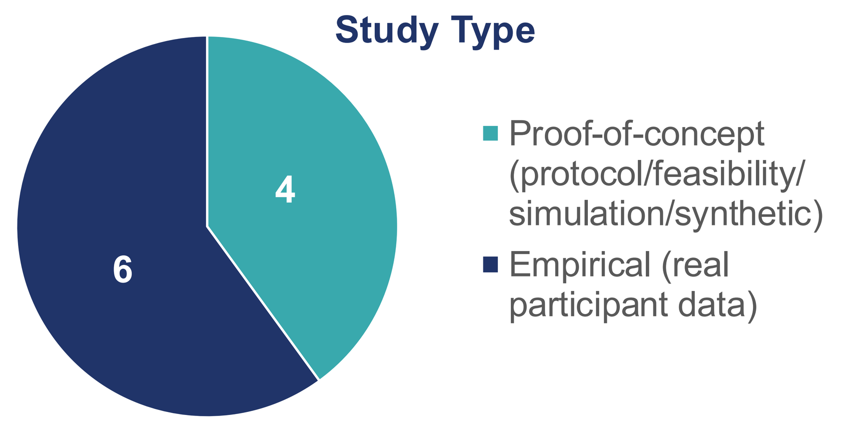
*Studies with multiple methods counted in each relevant category; total exceeds n=10

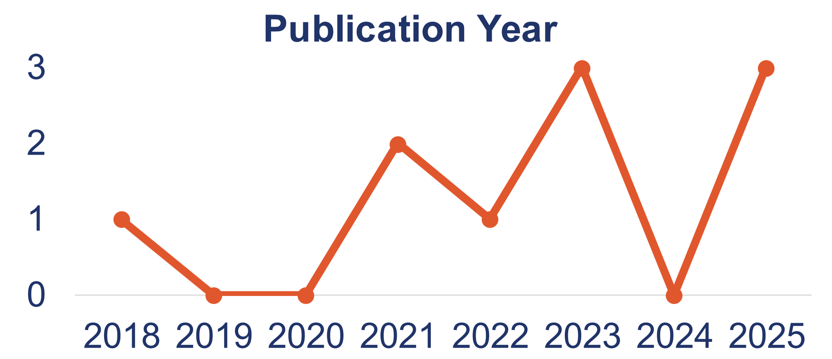
Computational Approach*



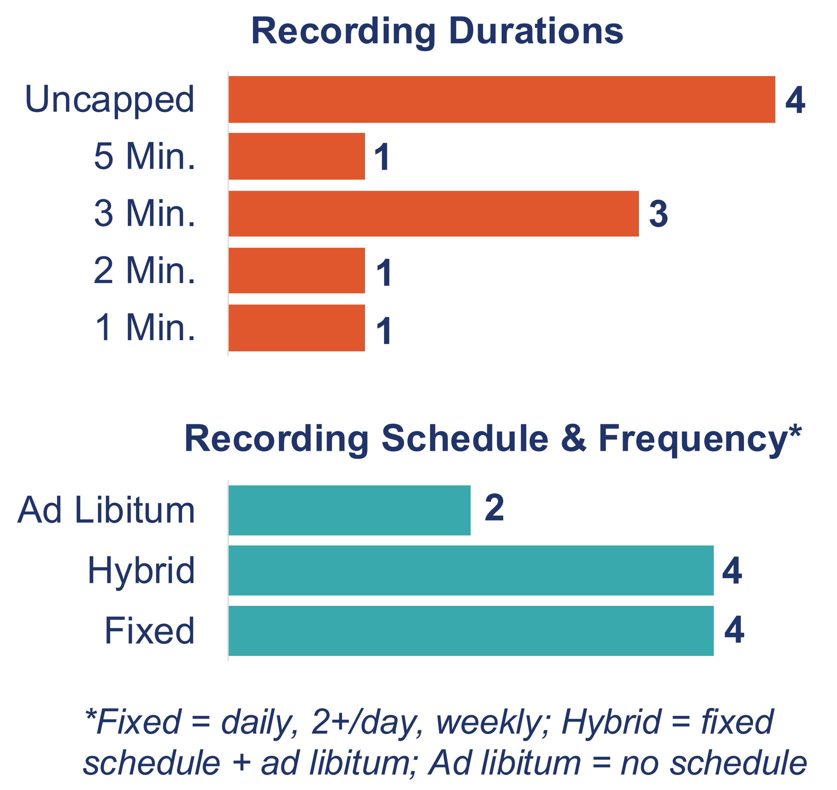
*Studies with multiple outcome types counted in each relevant category; total exceeds n=10

Target Outcomes*



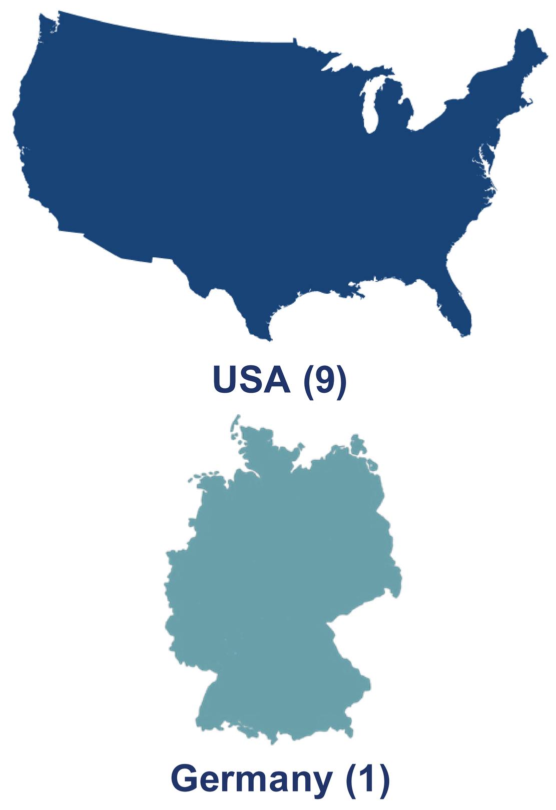


Field Nascency



*Fixed = daily, 2+/day, weekly; Hybrid = fixed schedule + ad libitum; Ad libitum = no schedule

Recording Protocol



Country of Studies

Gap 1

Signal underutilization
Most studies discard acoustic and paralinguistic channels, limiting biomarker discovery.

Gap 2

Lack of standardization
Heterogeneous protocols across voice diary frequency, duration, and prompting limit replication.

Gap 3

Limited generalizability
Field remains early-stage: ongoing data collection, synthetic/simulated data, typically small samples.

The Path Forward

STANDARDIZE
Harmonized collection protocols across frequency, duration, and prompting methods

TRANSPARENCY
Transparent, reproducible, and open-source computational methods

DIVERSIFY
Larger & more diverse cohorts to encompass representative speech samples

VALIDATION
Systematic validation to enable reliable clinical translation

The technological infrastructure exists**, with this agenda, voice diaries can evolve from exploratory tools into **clinically meaningful, scalable instruments for digital health.

Want to learn more? Scan for the paper & protocol!

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1. Kaplan DM, Alvarez SJ, Palitsky R, Choi H, Clifford GD, Crozier M, Dunlop BW, Grant GH, Greenleaf MN, Johnson LM, Maples-Keller J. Fabla: A voice-based ecological assessment method for securely collecting spoken responses to researcher questions. Behav Res Methods. 2025 Aug;57(9):257, doi:10.3758/s13428-025-02777-1
2. Silva WJ, Lopes L, Galdino MK, Almeida AA. Voice acoustic parameters as predictors of depression. J Voice. 2 024 Jan;38(1):77-85, doi:10.1016/j.jvoice.2021.06.018
3. Nemeti, E, Park T, Bozkurt S. Voice Diaries as Digital Biomarkers: A Scoping Review of Computational Methods and Applications. OSF Registries. 2025. [2025-11-03]. [DOI]